

RootRotAI: A deep learning-based mobile and web application for dry root rot disease identification and assessment in chickpea under research environment

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Abstract

Dry Root Rot (DRR), caused by the soil-borne fungal pathogen *Macrophomina phaseolina*, is an emerging threat to global chickpea production. Screening large chickpea germplasm collections, mutant populations, and transgenic lines for DRR resistance remains challenging due to manual disease scoring, which is subjective, labor-intensive and often prone to rating bias. To address this, our laboratory has previously developed RootRotAI 2.0, a set of deep learning models developed using TensorFlow to support automated detection and assessment of DRR from images of chickpea collected under laboratory conditions. In the current study, we focus on the high-throughput deployment of these models by developing a mobile application, RootRotAI 2.2, and, a web application, RootRotAI 2.1, to facilitate a user-friendly interface and broader accessibility. These applications allow users to upload or capture images to automatically diagnose and assess DRR severity. RootRotAI 2.2 and 2.1 integrate two transformer-based multitask deep learning models now optimized and deployed via TensorFlow Lite for efficient on-device inference. These tools can be accessed via <https://github.com/scipdatabase/RootRotAI> and mobile app is available in play store. The application supports three imaging modalities, a handheld camera, a root scanner, and a light microscope each with modality-specific backgrounds. These tools offer a real-time assessment capabilities through an offline mobile mode and a high-capacity platform for large-scale data analysis via the online web application. While image-based disease detection tools are common, RootRotAI serves as a novel and key digital resource for the identification of DRR in chickpea, ultimately facilitating crop improvement programs and contributes to sustainable agriculture.

Keywords: disease diagnosis, severity assessment, laboratory screening, germplasm, application, machine learning

Dry Root Rot (DRR), caused by *Macrophomina phaseolina*, is a significant threat to crop productivity worldwide and is further exacerbated by drought and heat stress (Chilakala et al. 2022; Chilakala et al. 2023; Irulappan et al. 2022). Identifying resistant sources within host germplasm collections and landraces that can tolerate DRR, particularly under concurrent drought stress, is critical for the discovery of novel donor lines for chickpea resistance breeding programs (Mirchandani et al. 2023; Rai et al. 2022). To address this, high-throughput screening methods have been developed, such as blotting paper technique using the agar plug method or dip inoculation (Durgadevi et al. 2025; Irulappan and Senthil-Kumar 2021). However, researchers often face significant challenges in accurately identifying and quantifying disease symptoms across large germplasm collections, transgenic lines and mutant populations evaluated across multiple replicates. The conventional method of disease scoring using manual observation is time-consuming, subjective, and often inconsistent, particularly during large-scale disease screening (Bock et al. 2022). Artificial intelligence (AI)-based approaches enable automated digital image analysis, improving accuracy and reproducibility over manual disease scoring methods (Murphy et al. 2024). This enables real-time disease assessment, valuable for research projects and repeated experimental evaluations. Consequently, there is a need for an accurate, scalable, and user-friendly disease scoring system suitable for both laboratory-based large-scale germplasm screening programs and individual studies. To facilitate accurate and automated disease scoring, our laboratory previously developed RootRotAI (Root Rot Assessment Interface) 2.0, that comprises a set of deep learning-based disease diagnosis and severity assessment models developed for automated DRR disease phenotyping in chickpea under laboratory conditions using blotting paper technique (Priya et al. 2026). These models process data from three distinct imaging modalities namely, a handheld camera, a root scanner and a light microscope that are commonly utilized in the laboratory environment. Handheld camera images provide a macroscopic view of the plant, allowing the model to detect overall symptoms. Root scanner images offer high-resolution, 2D projection of the root architecture, which are essential for quantifying necrosis. Microscope images capture necrotic cell death in the cellular level, enabling the identification of precise tissue damage.

In this announcement, we introduce the high-throughput deployment of RootRotAI for the identification and assessment of DRR under research environments. These models are now deployed via TensorFlow Lite as both a mobile application, RootRotAI 2.2, and a web application, RootRotAI 2.1, to facilitate broad adoption by breeders, and other researchers, and students. Both RootRotAI 2.2 and 2.1 utilize two multitask models based on the Vision Transformer (ViT) architecture. These models identify and classify DRR from healthy plants using a classification head and assess the disease severity using a standard linear regression head. The severity score is rounded to the nearest integer on a scale of 1-5, where a higher value indicates higher disease severity, whereas a score of 0 indicates a healthy plant. Classification accuracies of 94%, 85%, and 93%, and corresponding root mean square errors (RMSEs) of 0.87, 1.49, and 1.21 for disease severity scoring were obtained for the handheld camera, root scanner, and microscope modalities, respectively. This resource enables users to upload images and automatically diagnose DRR (Figure 1).

The mobile application, RootRotAI 2.2 was developed for the rapid, offline diagnosis of DRR with a user-friendly interface. To ensure compatibility with mobile environments, the original multitask models from Priya et al. (2026), which exceeded 200 MB, were quantized using TensorFlow Lite. This process successfully reduced the model sizes to 22.8 MB for the microscope modality and 23.3 MB for the handheld camera and root scanner modalities. Notably, this significant reduction in size had no discernible effects on performance, as model accuracy remained consistent (Table 1, Figure 2). This system was developed using the Flutter framework to maintain cross-platform efficiency, the applications allow users to select a modality from the dropdown menu and either upload images or capture them directly within the app (Figure 1b). To assist users, the application provides reference images of both healthy and DRR-infected plants for each of the three image categories. The application requires approximately five to ten seconds to load and two to three seconds to analyze each image. Results can be viewed within the app or downloaded as a CSV file, accessible in the "Downloads" section of the mobile device (Figure 1b). The application first employs a valid/invalid classifier to pre-filter images, ensuring that only those containing plant subjects are passed to the primary model for analysis. This is achieved through a simple CNN classifier utilizing the MobileNetV2 architecture with weights pre-trained on ImageNet images (Deng et al. 2009; Sandler et al. 2018). This

classifier was trained using valid datasets from all three modalities and invalid dataset, a subset of the ImageNet dataset consisting of 1000 non-plant images across various categories. When a user uploads the images, the app first converts them into .png format. Image preprocessing for valid vs. invalid classification and the DRR classification models are executed simultaneously in isolated memory heaps created with Flutter isolates. These are spawned temporarily for each image and are terminated to free up RAM on the user's local device. Once invalid images are filtered out, the app predicts the class and severity score using the lite versions of the models (Priya et al. 2026). The downloadable Android mobile application is 202 MB and is currently being used by 18 institutes across India. The GitHub release of the application is available at <https://github.com/scipdatabase/RootRotAI/releases/tag/v2.2>.

RootRotAI 2.1 is an online web application that provides all the functionalities available in the mobile version (Figure 1c). Unlike the mobile app, the web application is not constrained by the RAM of the user's local device, allowing the user to analyze large image datasets in a single session. However, a stable internet connection is required to minimize latency. The web application utilizes FastAPI to bridge the HTML frontend and the Python backend. The application is available via GitHub at <https://github.com/scipdatabase/RootRotAI>.

Instructions for Use

To obtain the best performance from RootRotAI 2.1 and 2.2, users must follow standard laboratory protocols using a pure culture of *M. phaseolina* inoculum to generate DRR. Following disease development, users can employ any one of the three modalities to capture images. Images must be captured in the designated formats, ensuring that handheld camera images utilize a red background, root scanner images utilize a blue background, and microscope images utilize a white background. The application supports all common image file types, including .png, .jpg, .jpeg, and .tif. It is essential that all images remain clear of any external objects to prevent interference with the analysis. Furthermore, users should ensure the plant is not affected by any disease other than DRR to maintain scoring accuracy. While the system is optimized for individual assessments, the handheld camera modality is the only format that allows for the inclusion of multiple plants within a single image. Users

are cautioned that any image that fails to meet these specifications, or uses random imagery, will be flagged as invalid.

In conclusion, while generalized disease prediction tools are widely available, RootRotAI serves as a specialized and novel digital resource tailored for DRR disease prediction and assessment in chickpea under research environments. By offering both mobile and web-based interfaces, this platform supports high-throughput deployment of automated phenotyping, enabling researches to analyze large-scale datasets with precision in a short period of time. These applications directly facilitate resistant breeding and crop improvement programs by streamlining the identification and deployment of resistant donor lines. All source codes, optimized models and deployment frameworks for RootRotAI 2.1 and 2.2 are accessible at <https://github.com/scipdatabase/RootRotAI>. The mobile app is available in play store (https://play.google.com/store/apps/details?id=com.rootrot_ai.app).

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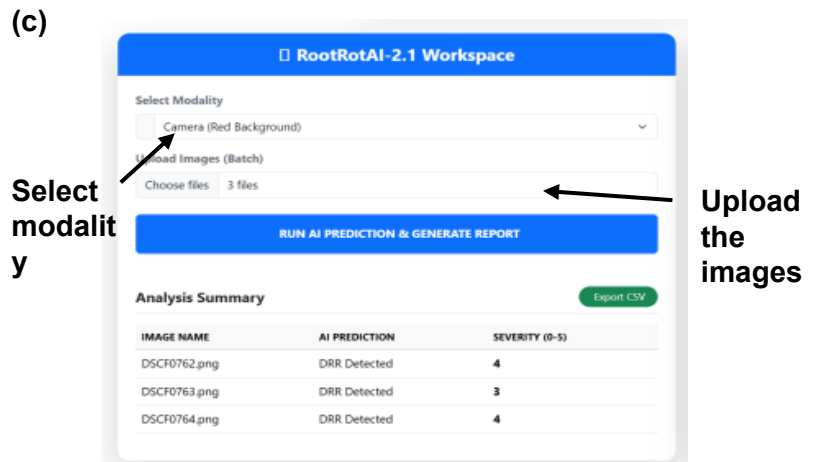
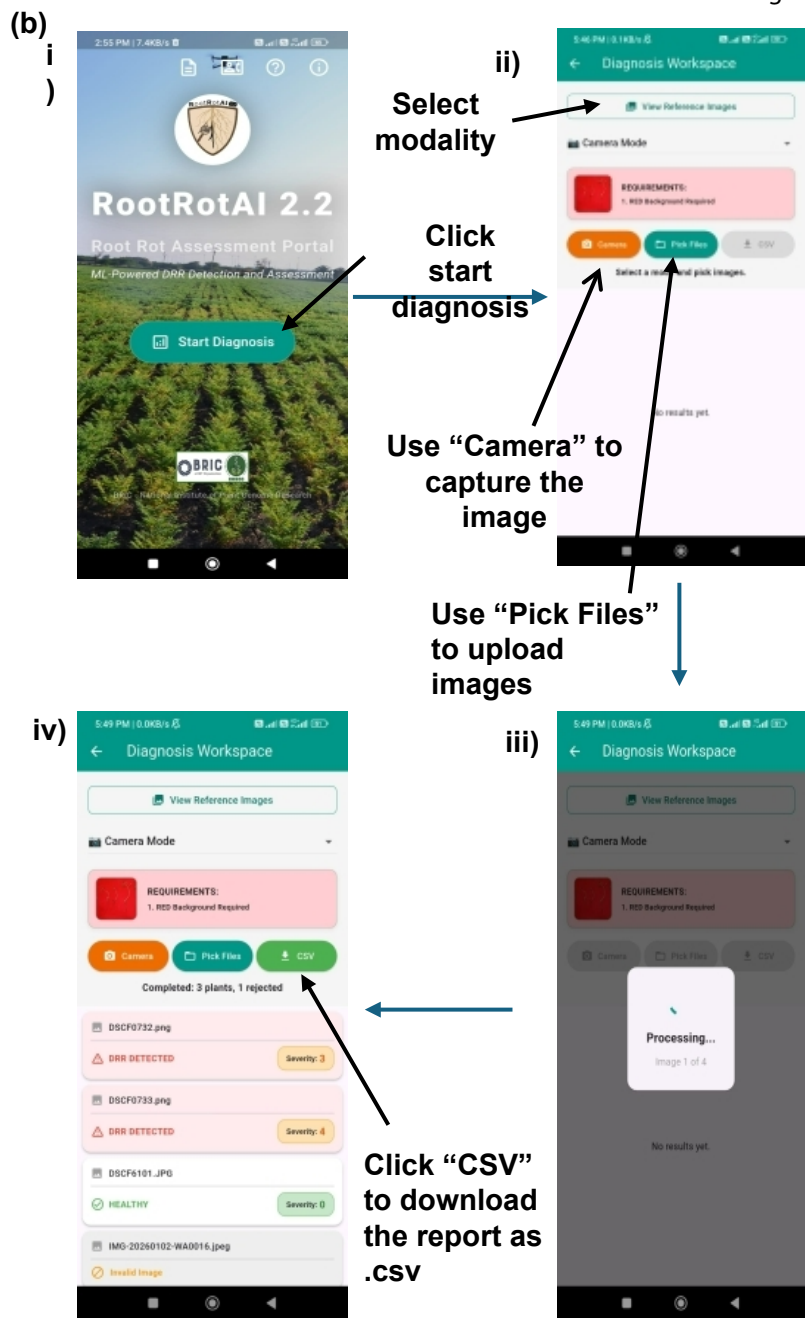
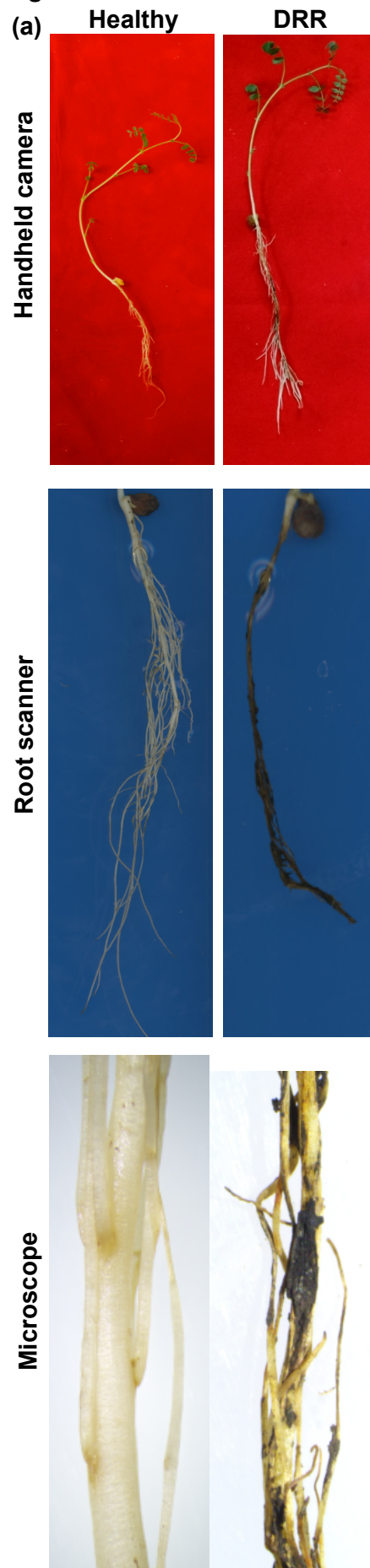
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Fig. 1. A step-by-step workflow of RootRotAI. The applications allow the user to upload images and automatically diagnose DRR and assess disease severity. (a) Representative images from the laboratory environment using the blotting paper technique for the handheld camera, root scanner, and microscope modalities for control and DRR. (b) A detailed workflow for RootRotAI 2.2 mobile application, depicting user navigation and key features. The interface snapshots include the (i) home page, (ii) diagnosis workspace, (iii) processing window, and (iv) results window of the mobile application. To initiate the process, the user selects ‘start diagnosis’ to navigate to the diagnosis workspace, where they choose a specific modality and either upload or capture the images. Once the images are processed, the application automatically displays the results window. (c) User interface and features of RootRotAI 2.1.

Fig. 2. Confusion matrices showing the performance of the mobile and web applications. Images were collected following standard imaging protocols after disease development. (a) Classification performance for valid vs. invalid image types for filtering non-plant images or those not meeting specified criteria. A total of 408 images were used for testing, out of which 200 were valid, and 208 were invalid. (b–d) DRR vs. control (healthy) classification performance for (b) the handheld camera modality (361 DRR and 361 healthy images), (c) the root scanner modality (224 DRR and 224 healthy images), and (d) the microscope modality (296 DRR and 296 healthy images).

Table 1. Evaluation metrics across modalities

Fig. 1.



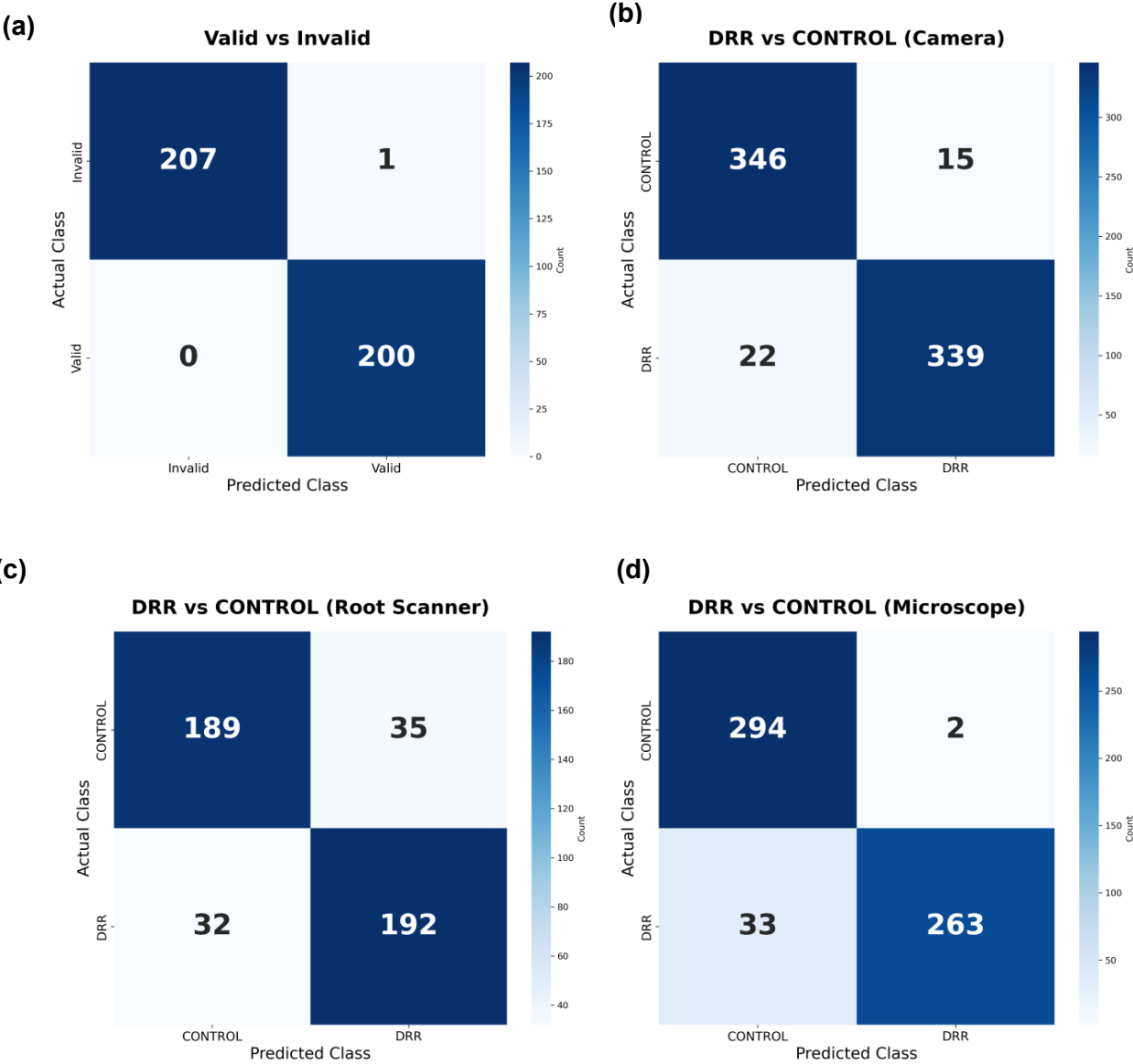


Table 1. Evaluation metrics across modalities

Model	Accuracy for classification	RMSE for severity score
Handheld camera	94%	0.87
Root scanner	85%	1.49
Microscope	93%	1.21